

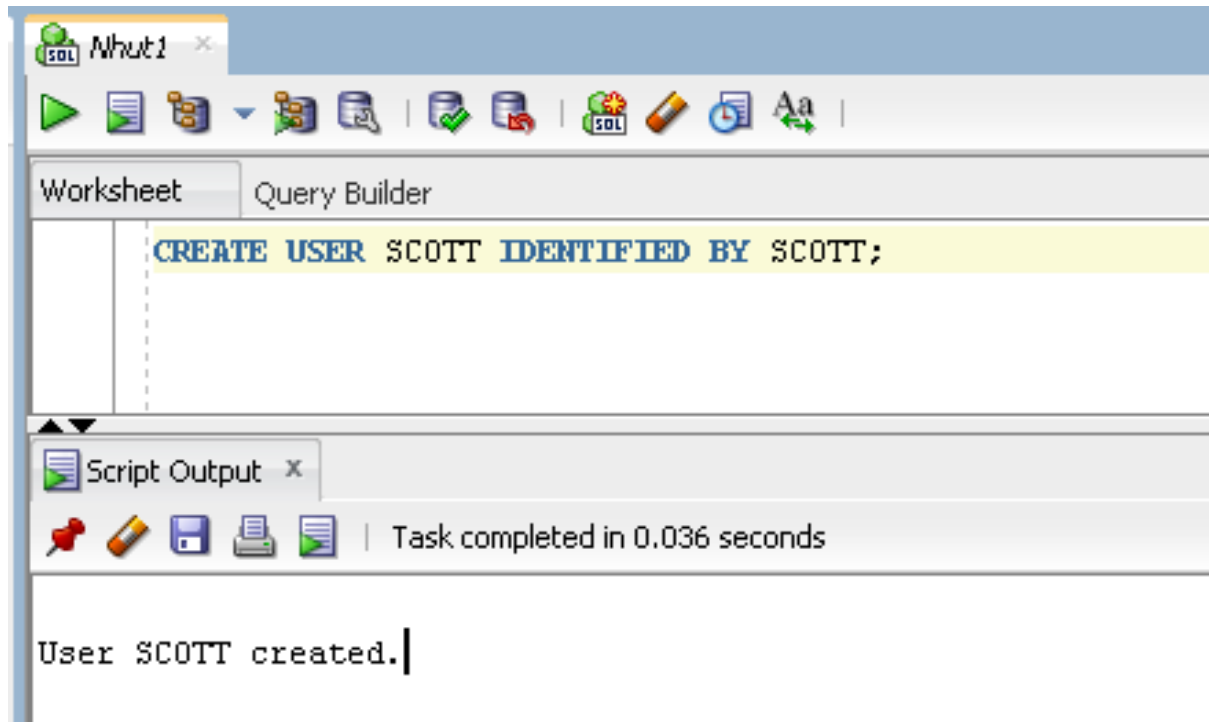
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Lab 5. User management and security

1. Create a user named SCOTT with the password as the same username.

Code:

```
CREATE USER SCOTT IDENTIFIED BY SCOTT;
```



2. List all users in the system.

(Hint: using dba_user view)

Code:

```
SELECT * FROM dba_users;
```

nhut2

Worksheet Query Builder

`SELECT * FROM dba_users;`

Query Result x

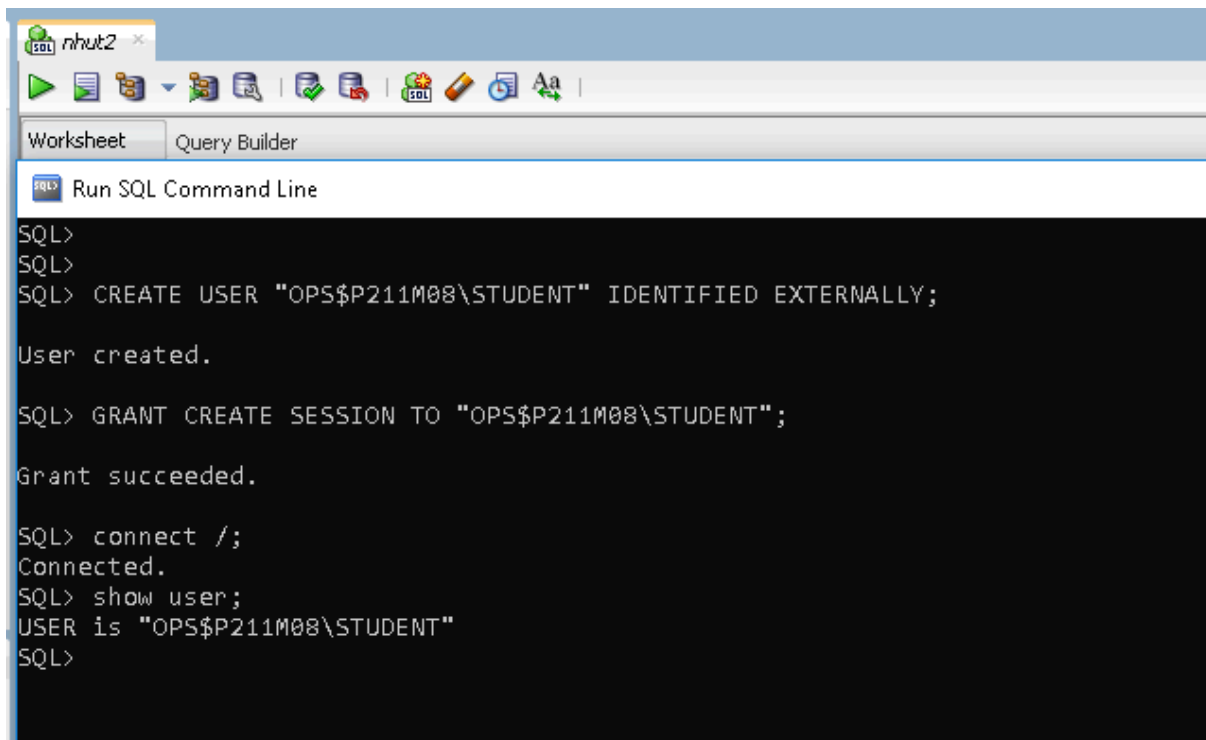
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	USERNAME	USER_ID	PASSWORD	ACCOUNT_STATUS	LOCK_DATE	EXPIRY_DATE	DEFAULT_TABLESPACE	TEMPORARY
1	SCOTT	49 (null)	OPEN		(null)	07-MAY-25	SYSTEM	TEMP
2	USER1	48 (null)	OPEN		(null)	07-MAY-25	SYSTEM	TEMP
3	SYS	0 (null)	OPEN		(null)	07-MAY-25	SYSTEM	TEMP
4	SYSTEM	5 (null)	OPEN		(null)	07-MAY-25	SYSTEM	TEMP
5	ANONYMOUS	35 (null)	OPEN		(null)	25-NOV-14	SYSAUX	TEMP
6	HR	43 (null)	OPEN		(null)	07-MAY-25	USERS	TEMP
7	APEX_PUBLIC_USER	45 (null)	LOCKED		29-MAY-14	25-NOV-14	SYSTEM	TEMP
8	FLows_FILES	44 (null)	LOCKED		29-MAY-14	25-NOV-14	SYSAUX	TEMP
9	APEX_040000	47 (null)	LOCKED		29-MAY-14	25-NOV-14	SYSAUX	TEMP
10	OUTLN	9 (null)	EXPIRED & LOCKED		08-NOV-24	08-NOV-24	SYSTEM	TEMP
11	DIP	14 (null)	EXPIRED & LOCKED		29-MAY-14	29-MAY-14	SYSTEM	TEMP
12	ORACLE_OCM	21 (null)	EXPIRED & LOCKED		29-MAY-14	29-MAY-14	SYSTEM	TEMP
13	XS\$NULL	2147483638 (null)	EXPIRED & LOCKED		29-MAY-14	29-MAY-14	SYSTEM	TEMP
14	MDSYS	42 (null)	EXPIRED & LOCKED		29-MAY-14	08-NOV-24	SYSAUX	TEMP
15	CTXSYS	32 (null)	EXPIRED & LOCKED		08-NOV-24	08-NOV-24	SYSAUX	TEMP
16	DBSNMP	29 (null)	EXPIRED & LOCKED		29-MAY-14	29-MAY-14	SYSAUX	TEMP
17	XDB	34 (null)	EXPIRED & LOCKED		29-MAY-14	29-MAY-14	SYSAUX	TEMP
18	APPOSSYS	30 (null)	EXPIRED & LOCKED		29-MAY-14	29-MAY-14	SYSAUX	TEMP

3. Create an external user (operating system) and connect using that user.
 (Hint: A user needs the *CREATE SESSION* privilege to connect to the Oracle DB)

Code:

```
CREATE USER "OPSP211M08\STUDENT" IDENTIFIED EXTERNALLY;
GRANT CREATE SESSION TO "OPSP211M08\STUDENT";
```



The screenshot shows a window titled 'nhut2' with a toolbar and tabs for 'Worksheet' and 'Query Builder'. Below the tabs is a button labeled 'Run SQL Command Line'. The main area is a black terminal window with the following text:

```
SQL>
SQL>
SQL> CREATE USER "OPS$P211M08\STUDENT" IDENTIFIED EXTERNALLY;

User created.

SQL> GRANT CREATE SESSION TO "OPS$P211M08\STUDENT";

Grant succeeded.

SQL> connect /;
Connected.
SQL> show user;
USER is "OPS$P211M08\STUDENT"
SQL>
```

Connect:

connect / ;

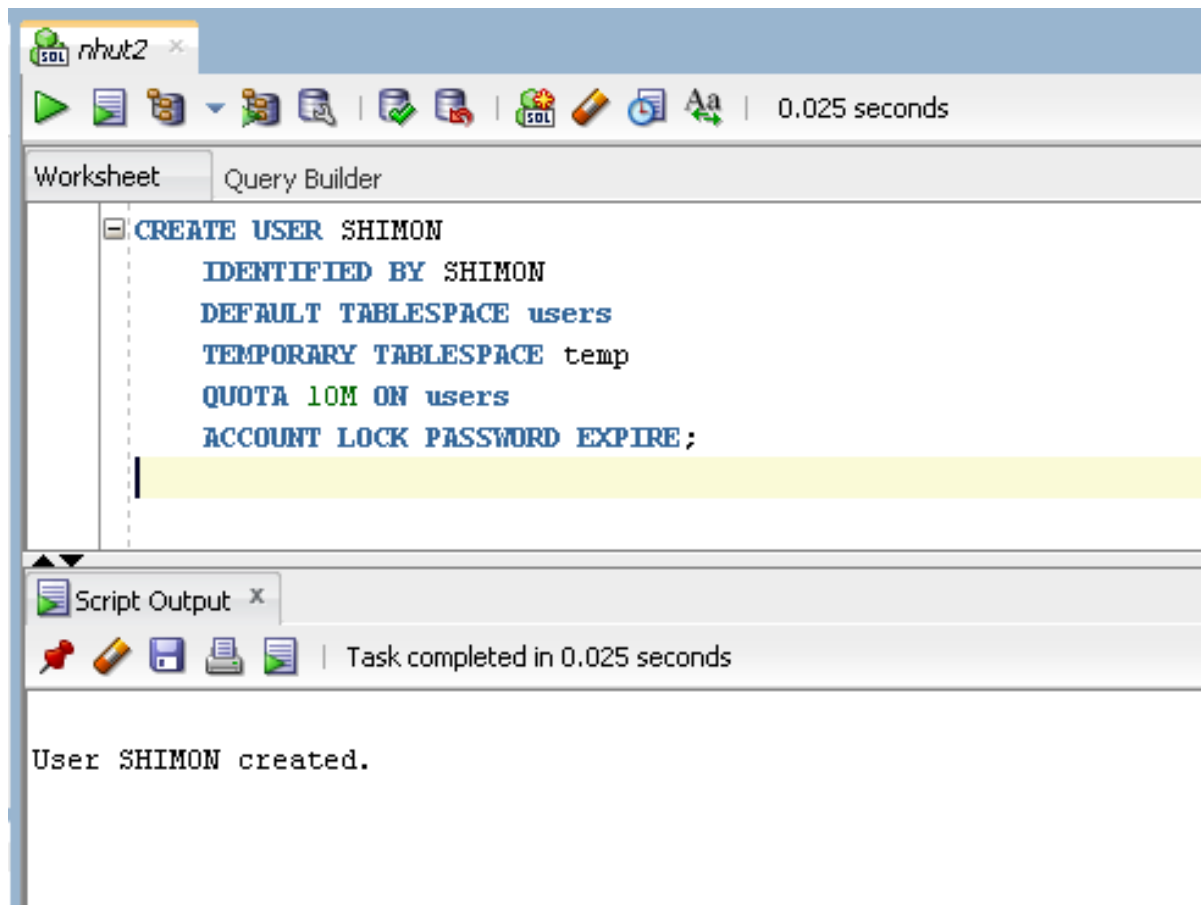
show user;

4. Create a user SHIMON with the following options:

- o Password: SHIMON
- o Default tablespace: users
- o Temporary tablespace: temp
- o Quota on the users tablespace: 10MB
- o Quota on example tablespace: unlimited
- o Account status: lock and expired

Code:

```
CREATE USER SHIMON
  IDENTIFIED BY SHIMON
  DEFAULT TABLESPACE users
  TEMPORARY TABLESPACE temp
  QUOTA 10M ON users
  ACCOUNT LOCK PASSWORD EXPIRE;
```

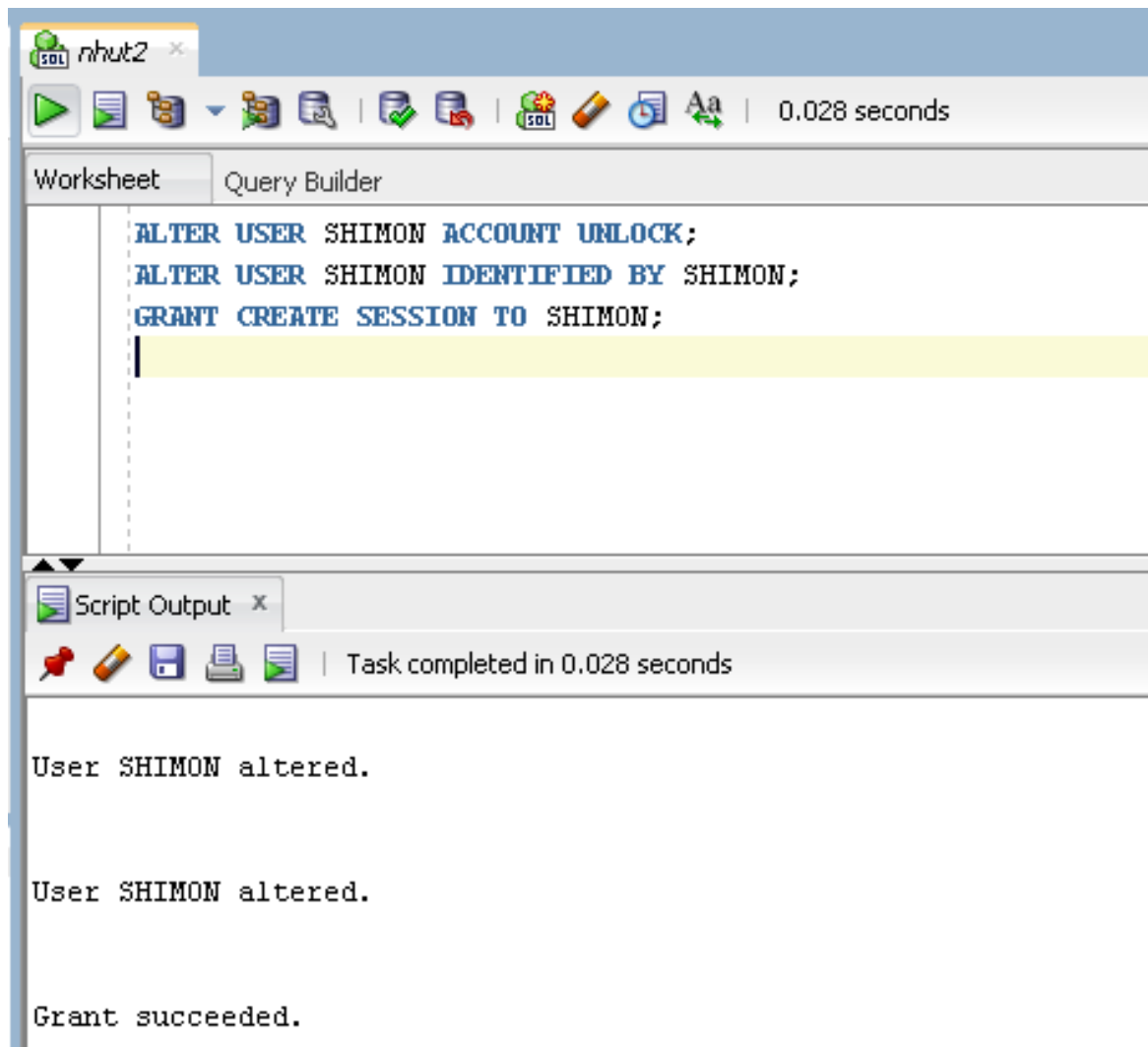


5. Connect to the database using the SHIMON account.

(Hint: You must unlock and grant the CREATE SESSION to the user SHIMON before connecting)

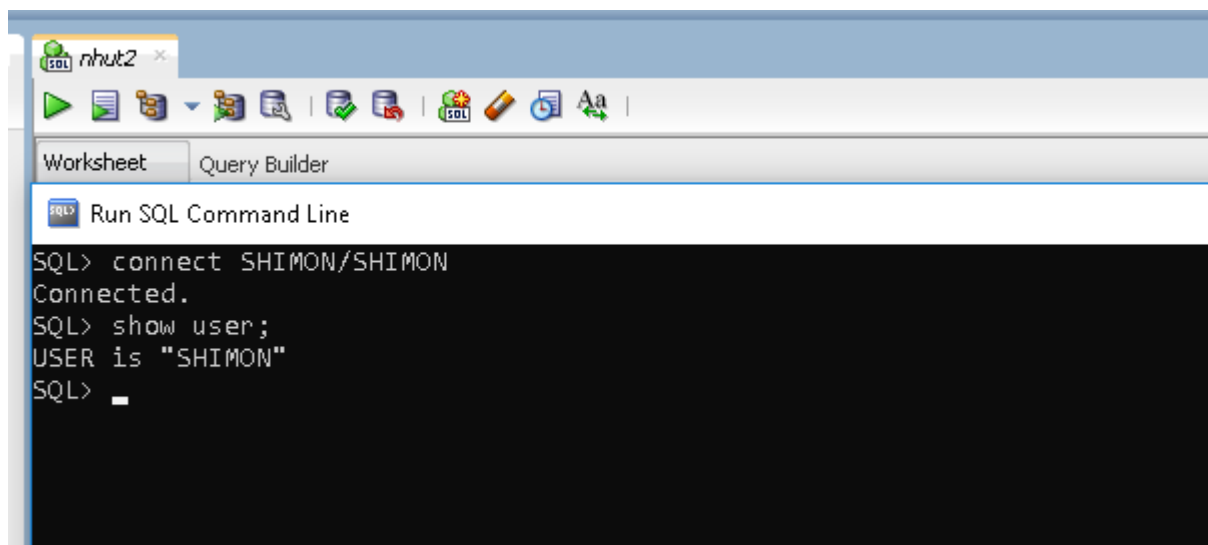
Code:

```
ALTER USER SHIMON ACCOUNT UNLOCK;
ALTER USER SHIMON IDENTIFIED BY SHIMON;
GRANT CREATE SESSION TO SHIMON;
```



Connect:

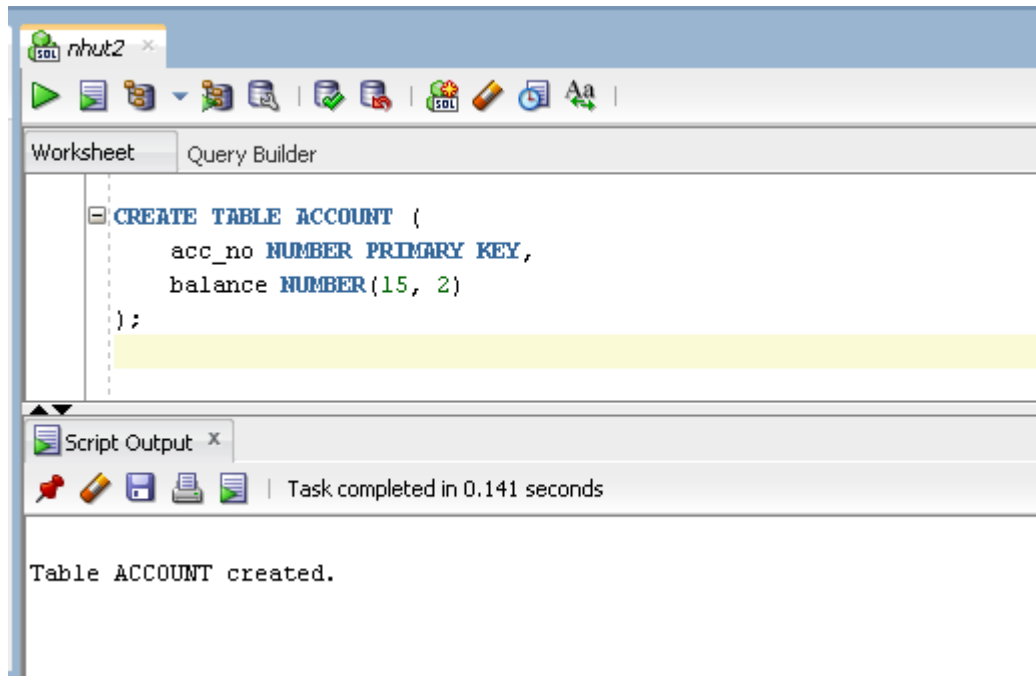
connect SHIMON/SIMON;



6. Create a table ACCOUNT with two fields: acc_no (primary key) and balance.

Code:

```
CREATE TABLE ACCOUNT (  
    acc_no NUMBER PRIMARY KEY,  
    balance NUMBER(15, 2)  
);
```



7. Create a role BASIC with the following privileges and grant this role to the user SHIMON:

```
CREATE TABLE  
RESOURCE  
DROP ANY TABLE  
UPDATE on the table HR.DEPARTMENTS
```

Code:

```
CREATE ROLE BASIC;
```

```
GRANT CREATE TABLE TO BASIC;  
GRANT RESOURCE TO BASIC;  
GRANT DROP ANY TABLE TO BASIC;  
GRANT UPDATE ON HR.DEPARTMENTS TO BASIC;
```

```
GRANT BASIC TO SHIMON;
```

nhut2 x

Worksheet Query Builder

```
CREATE ROLE BASIC;  
  
GRANT CREATE TABLE TO BASIC;  
GRANT RESOURCE TO BASIC;  
GRANT DROP ANY TABLE TO BASIC;  
GRANT UPDATE ON HR.DEPARTMENTS TO BASIC;  
  
GRANT BASIC TO SHIMON;
```

Script Output x

Task completed in 0.015 seconds

Role BASIC created.

Grant succeeded.

Grant succeeded.

Grant succeeded.

Grant succeeded.

Grant succeeded.

nhut2 x

Worksheet Query Builder

```
GRANT BASIC TO SHIMON;  
SELECT * FROM dba_role_privs WHERE grantee = 'SHIMON';
```

Script Output x Query Result x

All Rows Fetched: 1 in 0.004 seconds

	GRANTEE	GRANTED_ROLE	ADMIN_OPTION	DEFAULT_ROLE
1	SHIMON	BASIC	NO	YES

8. Create a profile *dev_prof* with the following resource limits and change the profile of user SHIMON to the *dev_prof* profile:

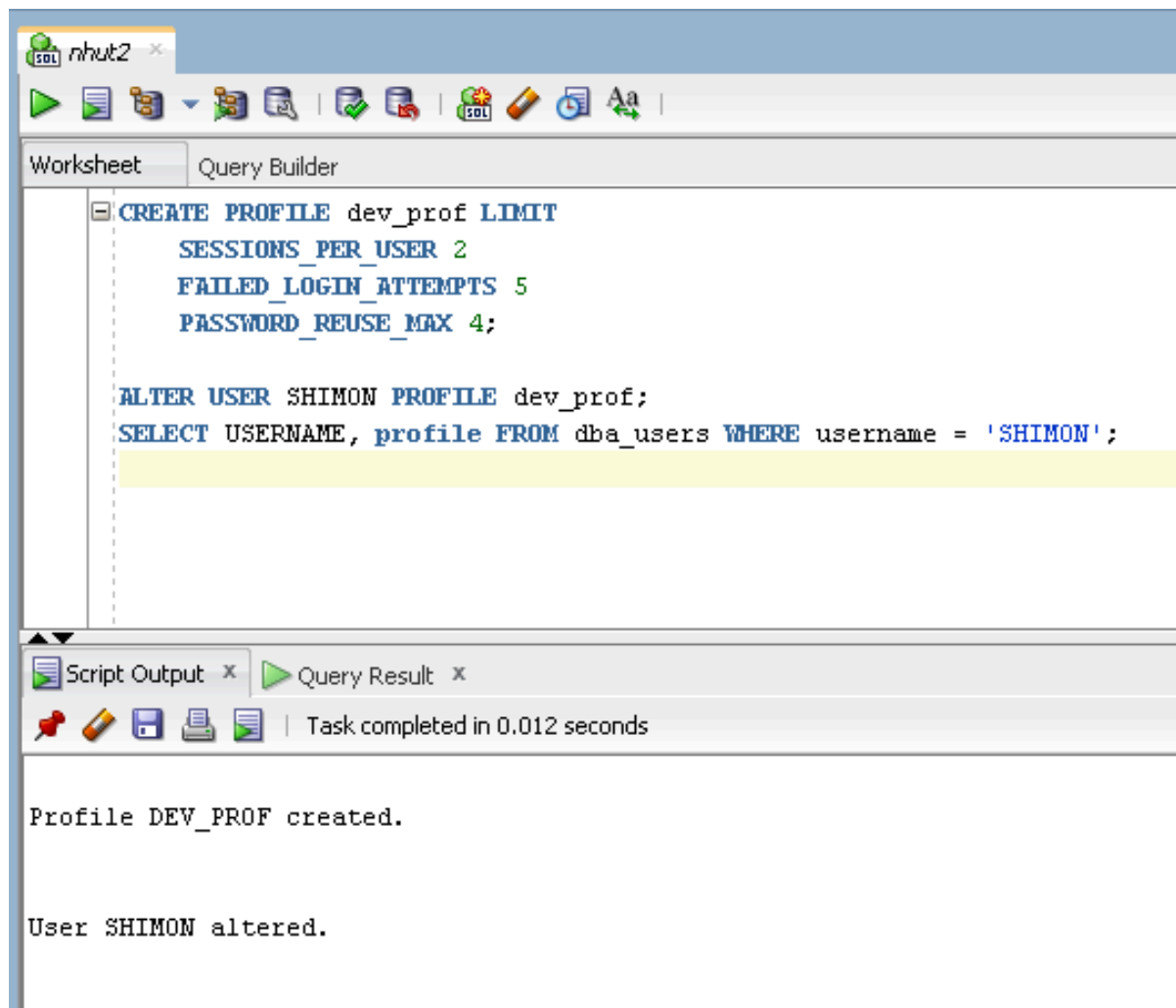
- A user can only open a maximum of 2 sessions at the same time.
- If a user enters the wrong password 5 consecutive times, their account will be locked.
- A user must change at least 4 different passwords before they can reuse a previously used password.

Code:

```
CREATE PROFILE dev_prof LIMIT  
  SESSIONS_PER_USER 2  
  FAILED_LOGIN_ATTEMPTS 5  
  PASSWORD_REUSE_MAX 4;
```

Test:

```
ALTER USER SHIMON PROFILE dev_prof;  
SELECT USERNAME, profile FROM dba_users WHERE username = 'SHIMON';
```



nhut2

Worksheet Query Builder

```
CREATE PROFILE dev_prof LIMIT
SESSIONS_PER_USER 2
FAILED_LOGIN_ATTEMPTS 5
PASSWORD_REUSE_MAX 4;

ALTER USER SHIMON PROFILE dev_prof;
SELECT USERNAME, profile FROM dba_users WHERE username = 'SHIMON';
```

Script Output x Query Result x

SQL | All Rows Fetched: 1 in 0.005 seconds

	USERNAME	PROFILE
1	SHIMON	DEV_PROF